

IMPROVING RESOLUTION OF FLUORESCENT MICROSCOPY USING SPECKLE ILLUMINATION AND JOINT SPARSE RECOVERY

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ABSTRACT

A variety of far-field super-resolution microscopy techniques have been proposed recently by exploiting nonlinear phenomenon in optics or specialized photo-switchable probes. Here, we present a *linear* optics based super-resolution microscopy for standard probes and experimental protocols. Rather than exploiting the nonlinearity in optics or photo-switchable probes, we use a *nonlinear* reconstruction algorithm that exploits joint sparsity of fluorescent emission from multiple random illumination. To maximize the resolution improvement owing to the joint sparsity, spatially incoherent speckle illumination is used for conventional epi-fluorescence microscopy setup. Experimental results obtained from a nano-pattern and bio-samples stained with standard fluorescent probes along with the proposed method demonstrate that a resolution of up to 37nm can be achieved.

Index Terms— Super-resolution microscopy, joint sparse recovery, speckle illumination

1. INTRODUCTION

Fluorescence microscopy has become an invaluable tool in biology due to its specificity for cellular functions. During the last decade, extensive investigations have been performed to achieve super-resolution fluorescent imaging using far-field optics [1, 2, 3, 4]. For example, STED microscopy [1] generates a nanoscale effective point spread function (PSF) by quenching excited fluorophores. In saturated structured illumination microscopy (SSIM)[2], structured illumination extends the spatial bandwidth of the optical system using Fourier domain synthesis [2], thereby achieving nano-scale resolution. STORM and PALM [4, 5] can achieve nano-scale resolution by using photo-switchable probes. Despite the differences in implementation, all of these approaches exploit nonlinear optical phenomena to overcome the diffraction limit. However, considering the large inventory of standard fluorophores or fluorescent proteins and well-established

standard imaging protocols, a new nanoscopy architecture that can use standard experimental protocols and fluorescence probes should be developed. Recently, new super-resolution imaging techniques [6, 5, 7] have been developed for standard probes by exploiting photobleaching/blinking; however, it is often difficult to apply these techniques for standard experimental protocols where photobleaching/blinking should be avoided. Sparsity-based reconstruction [8, 9] have been also introduced for high-density data to reduce the acquisition time for STORM such as DAOSTORM[8] and CSSTORM [9].

In this paper, we show that a joint sparse recovery approach can be used to improve the resolution of the conventional fluorescent microscopy imaging systems, which can be used for improving or developing new type of super-resolution microscopy. More specifically we describe a novel super-resolution microscopy using speckle andom illumination and joint sparse recovery. Unlike STORM that exploits random *emission* from photo-switchable probes, the proposed method uses random *illumination* so that it can be used for any standard probes, even including non-fluorescent probes. To obtain the super-resolution image from the overlapped illumination, we exploit that the emission from fluorescent probes can be overlapped in multiple illumination snapshots, while assuming that the probes themselves are sparsely distributed. Using joint sparse recovery algorithm called multiple sparse Bayesian learning (M-SBL) [10], we demonstrate significant resolution improvements.

2. THEORY

The key idea to break the diffraction limit then comes from employing a *nonlinear* reconstruction method, in particular a joint sparse compressed sensing (CS) [11, 12], to exploit the spatio-temporal sparsity of fluorescent emission from multiple random illuminations.

Specifically, image intensity measured at the m -th image sensor (CCD or CMOS) pixel at time sample t from a single fluorescence molecule target at $\theta = (u, v)$ is given by $y_m(t) = a_m(\theta)x_m(t)$, where $a_m(\theta)$ is a detector PSF from the fluorophore to the sensor pixel, and $x_m(t)$ denotes a corresponding fluorescence emission signal. For a detector com-

Grant sponsor: Korea Science and Engineering Foundation, Grant number 2011-0000353. We would like to thank Seung-Wook Ryu, Dongmin Keum, Ki-Hun Jeong, and Chulhee Choi at KAIST for providing the specimens.

posed of M pixels, if photons from multiple fluorophores impinge on the detectors, the formulation can be generalized as

$$\mathbf{y}(t) = \sum_{n=1}^N \mathbf{a}(\boldsymbol{\theta}_n) x_n(t) b_n,$$

where $\mathbf{y}(t) = [y_1(t), y_2(t), \dots, y_M(t)]^T$ is detector reading at time t , $\boldsymbol{\theta}_n$ denotes the position in any candidate location, $\mathbf{a}(\boldsymbol{\theta}) = [a_1(\boldsymbol{\theta}), a_2(\boldsymbol{\theta}), \dots, a_M(\boldsymbol{\theta})]^T$; and b_n is a binary flag to indicate the presence of a fluorescence probe at $\boldsymbol{\theta}_n$. Note that multiple illuminations change fluorescence emission temporal profile $x_n(t)$, whereas b_n does not vary. This implies that for all t , the illumination-emission product $x_n(t)b_n$ has nonzero values only on positions where $b_n = 1$, so it has the same nonzero spatial support. This property is called *jointly sparsity* in CS literature [11, 12]. For measurements from L random illumination, the joint sparsity can be stated as the number of nonzero row of matrix X is sparse, where $X = [x_n(t_l)b_n]_{n,l=1}^{N,L} \in \mathbb{R}^{N \times L}$.

In compressed sensing literature, joint sparse recovery is often called a multiple measurement vector (MMV) problem [11], and formulated as

$$\text{minimize } \|X\|_0 \quad \text{subject to } \|Y - AX\|_F < \epsilon,$$

where $Y = [\mathbf{y}(t_1), \dots, \mathbf{y}(t_L)]$, $A = [\mathbf{a}(\boldsymbol{\theta}_1), \dots, \mathbf{a}(\boldsymbol{\theta}_N)]$, and $\|X\|_0$ denotes the number of non-zero rows. In Eq. (1), $\|\cdot\|_F$ is Frobenius norm, and ϵ denotes the noise level. Theoretical analysis in [11] shows that $\text{rank}(X)$ needs to be maximized to take advantage of joint sparsity for the best resolution.

Since the non-zero position b_k does not vary, $\text{rank}(X)$ is determined by $x_n(t)$. Therefore, we exploit multiple random illumination to vary the temporal profile of fluorescent emission $x_n(t)$. In particular, we take advantages of one of the important properties of speckle statistics, spatial incoherence. More specifically, if spatially incoherent speckles are illuminated on fluorescence probes or proteins, we can easily expect that statistically nearly-uncorrelated fluorescence emission can be still obtained even with the spatial overlaps of the sources at specific time points. We are aware that due to illumination optics, the fluorescent emission is not perfectly uncorrelated. However, joint sparse CS does not need perfect uncorrelatedness since a full rank matrix X can be still obtained without requiring orthogonality between the rows.

In order to reconstruct the fluorescence emission matrix X using joint sparsity, we use M-SBL algorithm [10]. Under appropriate assumptions of noise and signal Gaussian statistics, one can show that M-SBL minimizes the following cost function in a so-called γ space [10]:

$$\mathcal{L}^\gamma(\gamma) = \text{Tr}(\Sigma_y^{-1} Y Y^*) + N \log |\Sigma_y| \quad (1)$$

where $\Sigma_y = \lambda I + A\Gamma A^*$ and $\Gamma = \text{diag}(\gamma)$. One of the most important contributions by Wipf is that the minimiza-

tion problem of the cost function (1) can be equivalently represented as the following standard sparse recovery framework [13]:

$$\min_X \mathcal{L}^X(X), \quad \mathcal{L}^X(X) = \|Y - AX\|_F^2 + \lambda g_{\text{msbl}}(X) \quad (2)$$

where $g_{\text{msbl}}(X)$ is a penalty given by

$$g_{\text{msbl}}(X) \equiv \min_{\gamma \geq 0} \text{Tr}(X^* \Gamma^{-1} X) + N \log |\lambda I + A\Gamma A^*|. \quad (3)$$

Wipf *et al* [13] proposed the following alternating minimization approach to solve the minimization problem (2). More specifically, for a given estimate $\gamma^{(t)}$ at the t -th iteration, we can find a close form solution for X in (2):

$$X^{(t)} = \Gamma^{(t)} A^* (\lambda I + A\Gamma^{(t)} A^*)^{-1} Y, \quad \Gamma^{(t)} = \text{diag}(\gamma^{(t)}).$$

Even for large problems, this can be easily computed using a standard conjugate gradient algorithm with an appropriate preconditioner. Then, for a given $X^{(t)}$ we need to solve the following minimization problem:

$$\gamma^{(t+1)} = \arg \min_{\gamma \geq 0} \mathcal{G}^{(t)}(\gamma)$$

where $\mathcal{G}^{(t)}(\gamma) \equiv \text{Tr}(X^{(t)*} \Gamma^{-1} X^{(t)}) + N \log |\Sigma_y|$. A fixed point update for γ is then given by

$$\gamma_i^{(t+1)} = \left(\frac{\frac{1}{N} \sum_j |x_{ij}^{(t)}|^2}{\mathbf{a}_i^H (\lambda I + A\Gamma^{(t)} A^*)^{-1} \mathbf{a}_i} \right)^{\frac{1}{2}}. \quad (4)$$

Note that the most computationally demanding part is the matrix inversion $(\lambda I + A\Gamma^{(t)} A^*)^{-1}$ since the image size is very large for microscopy imaging. To address this, the images are partitioned to overlapped blocks and M-SBL is applied for each overlapped block, and the final image is reconstructed as an average of the each M-SBL reconstructions.

3. EXPERIMENTAL RESULTS

An experimental setup was developed by modifying an illumination unit of a conventional efi-fluorescence microscopy. As shown in Fig. 1, light from 488nm Argon laser or 532nm diode laser is scattered by ground glass diffuser and then the scattered light is collimated by lens (f_2). The collimated and attenuated speckle pattern is then focused on the back focal plane of objective lens (100x, oil-immersion, NA 1.30, Olympus). Fluorescence emissions are detected using an EMCCD (Luca-S, Andor) through 1.6x relay lens. The EMCCD is synchronized with a step motor which rotates a glass diffuser at 10Hz frame rate.

For a quantitative analysis of the proposed method, we prepared a radial nano pattern as shown in Fig. 2. For the fabrication of the pattern, hydrogen silsesquioxane (HSQ, negative tone electron-beam resist) and Rhodamine 6G dyes

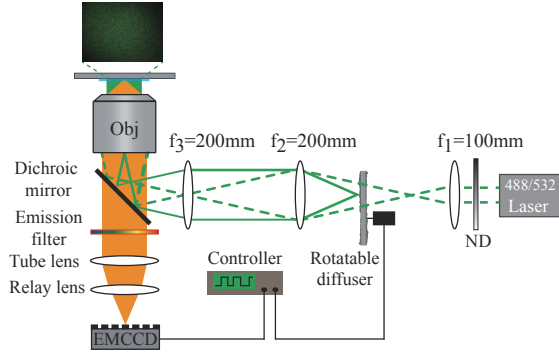


Fig. 1. An experimental setup. We generate speckle patterns from a rotatable diffuser illuminate a specimen through a 100x, 1.30NA, oil-immersion objective lens. The rotatable diffuser is synchronized with an EMCCD. Emission fluorescence signals from the sample are detected using an EMCCD through a 1.6x relay lens.

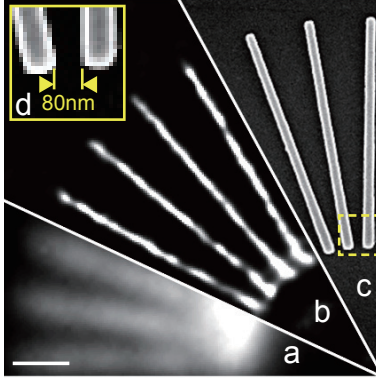


Fig. 2. Radial nano pattern with minimum distance of 80nm. (a) Diffraction limited image. (b) A reconstruction using the proposed method from 500 speckle illuminations. (c) Scanning electron microscope (SEM) image. (d) A close-up SEM image. Scale-bars are 500nm.

were mixed and then spin-coated on a glass substrate. Then, electron-beam lithography was used to form equi-angularly divergent lines whose width and height are 100nm, with minimum distance of 80nm as shown in Fig. 2.(d). We took 500 random speckle images using a 532 diode laser. As for joint sparse recovery algorithm, we used the M-SBL (multiple sparse Bayesian learning) algorithm due to its proven effectiveness for ill-posed sparse recovery problems [10]. In addition, we implemented a M-SBL using GPU(Nvidia Tesla C1060), so the reconstruction time for the proposed method was about 40sec for $1.5\mu\text{m} \times 1.5\mu\text{m}$ field-of-view. Compared with a conventional wide-field image in Fig. 2(a), a reconstruction using the proposed method in Fig. 2(b) shows clearly resolved line patterns, which well matches the pattern in SEM image in Fig. 2(c)

Next, sub-cellular structures such as microtubule and en-

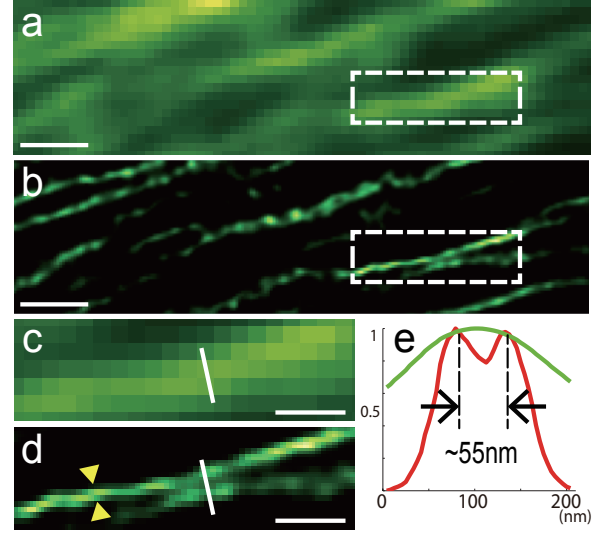


Fig. 3. Microtubule structure in HeLa cell stained by a fluorescent dye (Alexa 488). (a) Diffraction limited image, (b) reconstruction using the proposed method, (c,d) Close-up images of two dotted line boxes in (a,b), respectively. The FWHM in the yellow arrow in (d) is 37nm. (e) Line profiles of dotted line in (c,d). Green line is from a diffraction-limited image, and the red line is from the proposed method. Scale-bars are 500nm in (a,b) and 250nm in (c,d).

doplasmic reticulum (ER) in HeLa cell were imaged with 488nm argon laser as shown in Fig. 3. and Fig. 4. Microtubules are rope-shaped polymers of tublins, and a diameter of the microtubule is known as 25nm. Microtubules were stained by Alexa488 and 400 speckle images were taken. As shown in Figs. 3(a)(b), microtubule structure is clearly resolved in the proposed method compared to a diffraction limited conventional image. In close-up images Figs. 3(c)(d), the proposed method exhibits 37nm full width half maximum (FWHM) resolution at yellow marker, and clearly resolves two adjacent microtubule lines 55nm apart. In Fig. 4, smooth ER which consists of complex tubular structure, was stained by TRITC and 400 speckle images were used for the proposed method. The proposed method successfully resolved the complex tubular structure.

4. DISCUSSION AND CONCLUSIONS

The proposed method has many unique advantages. First, unlike SIM/SSIM, a tight control of illumination patterns is not required, as randomness of illuminations is used rather than pre-designed structures. Second, unlike the quantum-dot imaging that uses the uncorrelatedness and non-Gaussianity of the fluorescence emission [14], the proposed method exploits the algebraic property of joint sparsity, which makes it successful regardless of underlying statistics. Third, a high power laser that the generates saturated emission is not

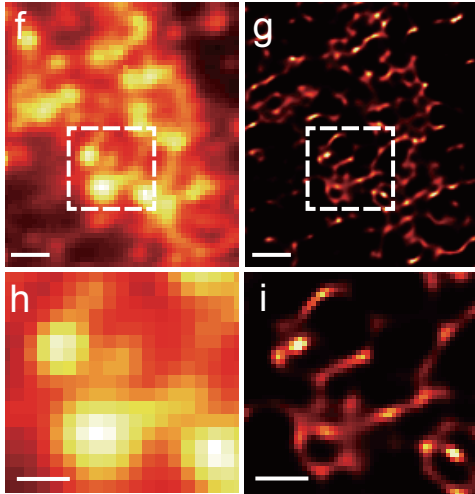


Fig. 4. ER in HeLa cell stained by a fluorescent dye (TRITC). (a) Diffraction limited image. (b) Reconstruction using the proposed method from 400 speckle illuminations. (c,d) Close-up images from dotted line boxes in (a,b) respectively. Scale-bars are 500nm in (a,b) and 250nm in (c,d).

needed, as the nonlinearity of the proposed method comes from a joint sparse CS rather than optical phenomenon. Accordingly, the proposed method has many unique advantageous in many standard bio-imaging experiments that requires avoidance of photobleaching.

In conclusion, we developed a novel far-field nanoscopy using random illumination and joint sparse compressed sensing recovery. To generate random illumination that maximizes the measurement diversity, quasi-random speckle patterns from a coherent source are used. A spatial resolution of up to 37nm was achieved using standard probes as well as standard experimental protocols.

5. REFERENCES

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